Experiment 10

Performance Optimization of the Vehicle Service Application

Aim

To optimize the performance of the Vehicle Service application in Zoho Creator by improving form loading speed, reducing unnecessary database operations, optimizing Deluge scripts and ensuring efficient report generation.

Procedure

1. Log in to Zoho Creator and open the Vehicle Service & Repair Request Management application.
2. Review the Service_Request_Form and remove unnecessary fields, scripts and workflows.
3. Optimize Deluge scripts by avoiding repeated database queries and unnecessary loops.
4. Use criteria-based record fetching to retrieve only the required service requests.
5. Optimize reports by displaying only essential columns and applying suitable filters.
6. Reduce unnecessary workflow executions and avoid duplicate email notifications.
7. Test the application after the changes to compare loading speed and responsiveness.
8. Verify that all service request features work correctly after optimization.

Optimization Actions

Area	Before	After
Form	Duplicate and unused fields (extra phone, unused media fields) on the form	Only required fields kept; fields grouped in sections A to H
Deluge fetch	Fetch all records, then test the status inside a loop	Criteria-based fetch: only Ready for Pickup records are fetched
Reports	All columns shown, no filters	Essential columns only; filters on Status and Service Type
Workflows	Notification fires on every edit, causing duplicate mails	Fires only when Status changes to Ready for Pickup

Deluge Optimization Example

BEFORE:

```
for each r in Service_Request_Form
{
    if(r.Service_Status == "Ready for Pickup")
    {
        sendmail
        [
            from : zoho.adminuserid
            to : r.Email_Address
            subject : "Vehicle Ready"
            message : "..."
        ]
    }
}
```

AFTER:

```
pending = Service_Request_Form[Service_Status == "Ready for Pickup" && Pickup_Notified == false];
for each r in pending
{
    sendmail
    [
        from : zoho.adminuserid
        to : r.Email_Address
        subject : "Vehicle Ready"
        message : "..."
    ]
    r.Pickup_Notified = true;
}
```

Process Diagram

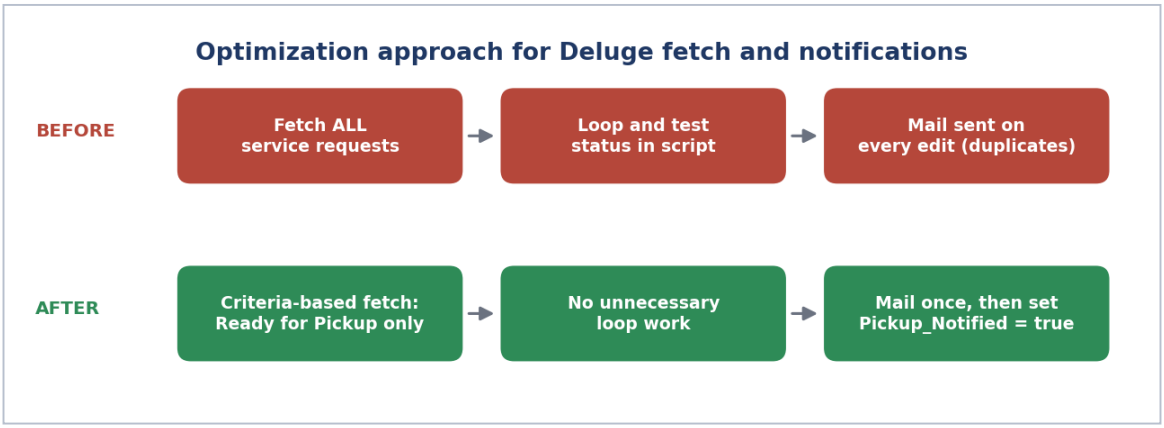


Figure 10.1: Diagram of the optimization approach (illustrative diagram, not a screenshot).

Before and After Comparison

Area	Before	After	Benefit
Form	Duplicate Vehicle Type field and unused fields	Only required fields, grouped in sections	Faster loading, fewer entry errors
Deluge fetch	All records fetched, tested inside a loop	Criteria-based fetch for Ready for Pickup	Less database work
Reports	All columns, no filters	Essential columns and filters	Quicker report display
Workflows	Mail on every edit	Mail only when Status becomes Ready for Pickup	No duplicate notifications

Verification Checklist After Optimization

Check	Expected Result
Submit a request	Record saved; acknowledgement mail sent once
Change status to Ready for Pickup	One notification; Pickup_Notified set to true
Open Service_Request_Report	Opens with essential columns; filters work
Kanban and Command Center page	Load and update correctly

Result

The Vehicle Service application was optimized by removing unnecessary fields, using criteria-based fetching, trimming report columns and reducing duplicate workflow executions, and all service request features were verified to work correctly after the changes.